Remote Sensing TEST

2025 C Division

Written by Amartya Das - Ward Melville

Team Name/ Number: _____

Competitor Names: _____

Tiebreakers: Questions in Reverse Order

SCORE: ____ / 128

Section 1: Acronyms [20 pts, 1 each]

Expand each of the following acronyms completely. Each question is worth one point, inexact answers will receive no credit.

1. GPS:
2. VTIR:
3. ASTER:
4. CZCS:
5. ARVI:
6. ADEOS:
7. CIESIN:
8. HSRL:
9. LVIS:
10. MBES:
11. MIMR:
12. TOLNet:
13. TSAVI:
14. WDVI:
15. TBM:
16. SSS:
17. RGB:
18. POEMS:
19. S-GCOS:

20. SaaS:

Section 2: Instrumentation & Physics [20 pts]

21. Explain the difference between passive and active sensors (2 pts):

22. For each of the following, classify it as passive or active (6 pts, 1 each)
    a. Radar
    b. Sonar
    c. Cameras
    d. Lidar
    e. Infrared Sensors
    f. Radiometers

23. Explain the difference between image quality and fidelity (2 pts)

24. How is the peak wavelength of emitted radiation related to the absolute temperature of the black body? (2 pts)

25. What wave property changes to cause refraction? (2 pts)

26. What is the gradual loss of force of a wave called? (2 pts)

27. What property does reflection off shiny surfaces have, and what is the technical name? (2 pts, one for property and one for name)

28. What word describes the movement of a wave from one medium to another? (2 pts)

Section 3: Interpretation [20 pts]

Refer to the image sheet for this question. For each of the five images, identify which satellite the image is taken from, and what the primary purpose of that satellite is. Each correct satellite name/ purpose will earn 2 points.

29.

30.

31.

32.

33.

Section 4: Climate Processes + Energy [20 pts, 2 each]

34. Which of the following is a primary characteristic of the El Niño phase of ENSO?

A) Unusually strong trade winds across the Pacific.

B) Warming of sea surface temperatures in the central and eastern equatorial Pacific.

C) Below-average rainfall in the southwestern United States.

D) Increased upwelling of cold, nutrient-rich waters off the coast of Peru.

35. During a La Niña event, what is typically observed in the eastern Pacific Ocean?

A) Weakening of the Walker Circulation.

B) Unusually cold sea surface temperatures.

C) An increase in hurricane activity in the eastern Pacific.

D) Reduced global precipitation.

36. The conversion of ammonia (NH_3) or ammonium (NH_4^+) into nitrites (NO_2^-) and then into nitrates (NO_3^-) by bacteria in the soil is known as:

A) Denitrification

B) Nitrogen fixation

C) Nitrification

D) Ammonification

37. Unlike the carbon and nitrogen cycles, the phosphorus cycle does not typically involve:

A) Biological absorption by plants.

B) Cycling through rocks and sediments.

C) Movement through aquatic ecosystems.

D) A significant atmospheric gaseous phase.

38. Which of the following represents the largest natural reservoir of carbon on Earth?

A) The atmosphere

B) Living biomass

C) The oceans

D) Sedimentary rocks and fossil fuels

39. What is the average albedo of the Earth?

40. Explain ice-albedo feedback.

41. What factors could lead to increased cloud albedo?

42. Describe a climate feedback loop other than ice-albedo feedback.

43. Explain the radiative forcing effects of aerosols.

Section 5: Mixed MCQ [20 pts, 2 each]

44. Which of the following statements are correct when describing the spectral signatures of vegetation and water?

    A) Water and vegetation reflect somewhat similarly in the visible wavelengths

    B) Similarity range is from about 0.4 to 0.7 nm but are almost always separable in the infrared

    C) Both of the above

    D) None of the above

45. ERS, Envisat, Sentinel and RISAT are example of which type of satellites:

    A) Optical

    B) Passive

    C) Thermal

    D) Microwave

46. At what wavelength will a planet with a surface temperature of 288 K radiate the most energy?

A) 45 nm

B) 11 μm

C) 53 nm

D) 90 μm

47. Which greenhouse gas is the primary cause of the breakdown of ozone?

A) Chlorofluorocarbon

B) Carbon dioxide

C) Methane

D) Nitrogen Trifluoride

48. Approximately what percent of the atmosphere is oxygen?

A) 78%

B) 21%

C) 26%

D) 1%

49. What color corresponds to the 380-450 nm range?

A) Red

B) Yellow

C) Green

D) Violet

50. What is the emissivity of a blackbody?

A) 1

B) 0.75

C) 0.25

D) 0

51. Cloud condensation nuclei are around what size?

A) 2 nanometers

B) 0.2 micrometers

C) 2 centimeters

D) 0.2 millimeters

52. Where are atmospheric CO2 levels measured?

A) French Alps

B) Key Largo

C) Mauna Loa

D) Maldives

53. Which of the following are false about atmospheric refraction?

A) Refraction in the atmosphere is caused by varying air densities at different heights

B) The refractive index of the atmosphere increases as altitude increases

C) Air pressure, temperature, and humidity are all factors that affect the atmosphere's refractivity

D) The refractive index of a layer in the atmosphere is the speed of light in a vacuum divided by the speed of light in that layer of the atmosphere

Section 5: Mixed FRQ [28 pts, 4 each]

54. Name and discuss at least two atmospheric or surface conditions that can lead to changes in equilibrium temperature of the Earth.

55. Relate the concentrations of one greenhouse gas and one trace gas to natural influencing factors.

56. Discuss an approach to combat climate change not related to reduce, reuse, recycle.

57. Discuss 3 different aspects of image quality, and how each contributes to data collection.

58. Discuss techniques used to identify the composition of atmospheric gas.

59. An ideal blackbody radiator has a peak emission wavelength of 5.0×10^{-7} m. Determine the surface temperature of this blackbody.

60. Consider the star Sirius A, which has a surface temperature of 9,667 degrees Celsius. Determine the total intensity of the star. Express your answer in $J/(s \cdot m^2)$.